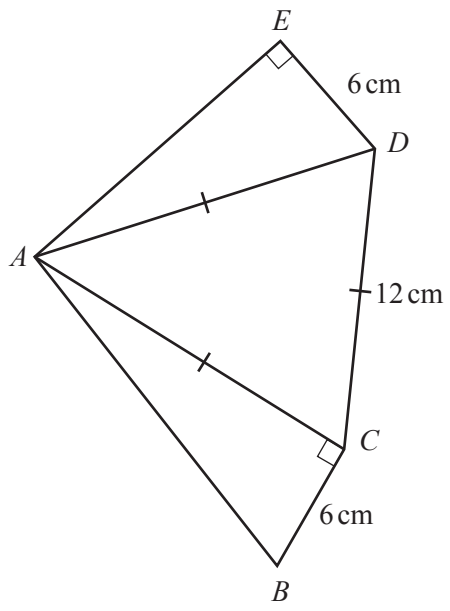
0580_s18_qp_43_Q6

6 (a)



NOT TO SCALE

In the pentagon *ABCDE*, angle *ACB* = angle *AED* = 90°.
Triangle *ACD* is equilateral with side length 12 cm.
DE = *BC* = 6 cm.

(i) Calculate angle *BAE*.

Angle *BAE* = ... [4]

(ii) Calculate *AB*.

AB = ... cm [2]

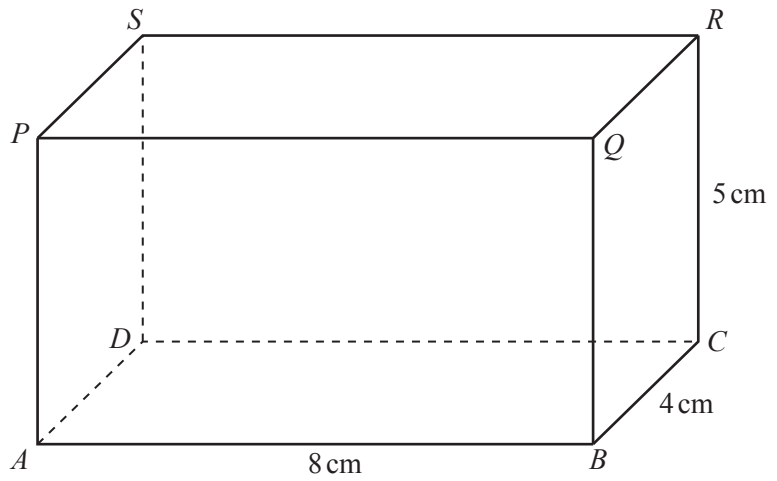
(iii) Calculate *AE*.

AE = ... cm [3]

(iv) Calculate the area of the pentagon.

..... cm^2 [4]

(b)



NOT TO SCALE

The diagram shows a cuboid.
$AB = 8$ cm, $BC = 4$ cm and $CR = 5$ cm.

(i) Write down the number of planes of symmetry of this cuboid.

..... [1]

(ii) Calculate the angle between the diagonal AR and the plane $BCRQ$.

..... [4]